

Console Connect by PCCW Global: Connecting multinationals, business partners, clouds, and apps all over the world

Tell us your challenge. We're here to help.

Contact us (/contact)

ABOUT PCCW GLOBAL

PCCW Global

About PCCW Global

PCCW Global (<https://www.pccwglobal.com> (<https://www.pccwglobal.com>)) is a leading communications service provider, offering mobility, voice, and data solutions to multinationals, telecommunications partners, cloud, and application service providers. It builds best-in-class connections integrated and orchestrated by Console Connect, its on-demand digital Software Defined Interconnection® platform, one of the first global platforms to fully automate switching and routing of all communications for seamless interconnection.

Google Cloud results

- Enables cloud-native containerized strategy to

Enabl
releas
week,

seamlessly
migrate more
than a million
lines of
application
code in
minute

- Eliminates
capital
expenditure
by running
on-premises
applications
on Google
Cloud since
day one
- Scales
automatically
to support
distributed
ledger
technology,
which
speeds up
settlements
from four
weeks to four
minutes
- Minimizes
application
downtime
from
monolithic
architecture

every
mont

Industries: Technology

Location: Australia

(<https://cloud.google.com/kubernetes-engine/>)



About Clear

Clear was founded by a team of seasoned entrepreneurs who wanted to remove friction in complex B2B trade by enhancing efficiency in the industry. The company builds blockchain-based settlement and clearing networks for global industries, while prioritizing data confidentiality, allowing smart B2B contracts to be created and executed automatically.

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Google Cloud (<https://cloud.google.com/>)

Kubernetes Engine
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The telecom industry is going through a digital transformation driven by changing customer behaviors. Consumers are used to buying anything at any time with a tap of a button in their everyday lives. Enterprise buyers are looking for the same convenience to purchase bandwidth online, and on demand

(<https://www.bain.com/insights/telcos-400-billion-as-a-service-enterprise-gold-mine/>)

, instead of being tied to a long-term contract.

PCCW Global (<https://www.pccwglobal.com/>), a leading global telecom provider, is expanding its ability to provide bandwidth services on demand, with the help of Google Cloud (<https://cloud.google.com/>). In 2018, PCCW Global launched the Console Connect (<https://www.consoleconnect.com/>) platform, hosted in Google Cloud, that automates and simplifies the bandwidth provisioning and settlement process through an easy-to-use web platform. Businesses and other carriers can order network as a service from an ecosystem of interconnect and cloud providers with one click.

“When a customer requests for connectivity on Console Connect, the critical components that orchestrate the process reside in Google Kubernetes Engine ([/kubernetes-engine](https://kubernetes-engine)),” says Paul Gampe, Chief Technology Officer at PCCW Global. “The actual physical infrastructure that’s automated is PCCW Global’s robust global network. Google Cloud helps us bring innovation to the market with this industry-first software defined cloud interconnection (SDCI) platform.”

As enterprise demand for data continues to rise, Console Connect makes it easy for customers to deliver their digital business anytime, anywhere, and only pay for what they use. For example, a content provider may use Console Connect to spin up bandwidth in seconds for a live broadcast to millions of viewers from a remote location, and scale it back down when the breaking news event is over.

“By running
microservices in
Google
Kubernetes
Engine, our
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side and
independently to
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What that
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—Paul Gampe, Chief
Technology Officer, PCCW
Global

Unlocking the power of containers to enable continuous development

Acquired by PCCW Global in 2017, the cloud-native platform was developed as microservices on Google Kubernetes Engine, enabling workload portability to other Kubernetes platforms across clouds and on-premises.

“When we were a venture capital-backed startup in Silicon Valley, the Console Connect platform lived in the US Google Cloud region,” says Paul. “Post-acquisition, we demonstrated the power of containers and Google Kubernetes Engine by seamlessly migrating the application, with its more than a million lines of code, to Asia in minutes. That’s only possible because of our cloud-native containerized strategy focused on Kubernetes.”

“By running microservices in Google Kubernetes Engine, our developers can innovate side by side and

independently to achieve continuous integration and development,” says Paul. “What that means for our customers is that we’re delivering features faster than we would have done in a large monolithic application, where there are too many dependencies that slow things down.”

“Consolidating on Google Cloud, instead of managing a mix of vendor environments, reduces our total cost of ownership. We now have engineers who can share knowledge and expertise with each other to manage our Google Cloud footprint.”

*—Paul Gampe, Chief
Technology Officer, PCCW
Global*

Prior to Google Cloud, PCCW Global managed a network of different cloud and on-premises IT systems, due to acquisitions over time. That meant the technical community was divided into multiple teams with specialized skills to support different platforms. PCCW Global migrated most of its on-premises applications to Google Cloud, using direct connectivity via Console Connect to maintain low latency and secure access between on-premises and cloud. Before migrating on-premises applications to the cloud, Paul considered these questions “Can I offload this VM into Google Cloud as pure OpEx? What cloud-native services can reduce IT operations cost? Can I scale it down when it's not in use and scale it up?”

With 20+ VMs deployed across multiple regions/zones, the ERP application is the most significant system that PCCW Global runs on Google Cloud. In migrating its ERP to the cloud, the company gains real-time operational and financial performance visibility to support business decisions. Using Cloud SQL to store and manage its financial data, PCCW Global can scale resources without investing in equipment that increases its fixed costs. Cloud Armor helps detect and block attempted exploits of CVE-2021-44228 and CVE-2021-45046 while the development team is still

patching the system. Application security, stability, and availability are improved with ease.

“Consolidating on Google Cloud, instead of managing a mix of vendor environments, reduces our total cost of ownership. We now have engineers who can share knowledge and expertise with each other to manage our Google Cloud footprint,” says Paul. “For example, an engineer in Athens can come online to solve a problem for an engineer based in Australia, and hand it over to an engineer in the United States.”

Cloud computing enables PCCW Global to optimize resources for more demanding workloads, such as threat management, which requires data ingestion and enrichment. Such workloads were previously hosted on dedicated servers.

Considering the cost of server provisioning and maintenance, it did not make sense for PCCW Global to move workloads around as they stood. The data science team found that by replatforming workloads into Google Cloud, and adopting services such as Cloud Functions, Pub/Sub, Cloud Dataproc and Google Cloud Storage in place of original scripts, Logstash, Spark and Hadoop components, the company was able to easily scale, maintain and observe ingestion and enrichment components.

With the help of Google Cloud Professional Services, PCCW Global optimizes cost and performance by determining the right components for different workloads. The team can now dynamically allocate

memory, storage, and processing resources
according to project needs.

“Distributed
ledger
technology can
be very compute
intensive and
variant in terms
of its load. An
advantage of our
cloud-native
approach is the
ability to scale
up infrastructure
on Google Cloud
for our
distributed
ledger
technology
partners.”

—Paul Gampe, Chief
Technology Officer, PCCW
Global

Creating value for customers through a vibrant ecosystem of partners

As a tech player, PCCW Global adopts an ecosystem view to build new capabilities and accelerate application innovation. While PCCW Global manages its own team of developers to create and deploy new services, it also works with a broad range of partners to quickly and efficiently scale up services.

When PCCW Global identifies viable use cases, it runs a proof of concept on Google Cloud to test the application for commercial readiness. To set it up, the team spins up a Kubernetes cluster using Google Kubernetes Engine and deploys the application in a test environment that replicates the production environment. For this project, it worked with distributed ledger technology partner [ClearX](https://www.clearx.io/) (<https://www.clearx.io/>).

“Now that we’ve solved the problem of automated connectivity, we’ve successfully developed a proof of concept with ClearX to automate intercarrier settlement with a distributed ledger solution on Google Cloud,” says Paul. “Distributed ledger technology can be compute intensive and variant in terms of its load. An advantage of our cloud-native approach is the ability to scale up infrastructure on Google Cloud for our distributed ledger technology partners.”

Transaction volumes are dramatically increasing as businesses continue to adopt new technologies such as automation, internet of things, and machine

learning to transform their businesses. In the past, a cloud service provider may procure 100 gigabits of capacity for a two-year contract. PCCW Global now receives requests from enterprise customers for 50 gigabits of capacity from Singapore to Sydney for two days.

“The traditional once a month billing routine doesn't work when you have highly scalable transaction rates,” says Paul. “Distributed ledger technology can speed up settlements from four weeks to four minutes by tracking concurrent multi-party transactions. We can potentially reduce transaction costs by using digital currencies such as stablecoin instead of third-party clearing houses.”

Supporting PCCW Global to enable business continuity for more customers

As a worldwide service provider, PCCW Global needs to ensure continuous and uninterrupted service availability and delivery to its customer base. During the COVID-19 pandemic in 2020, PCCW Global experienced increased demand from customers and partners to scale up connectivity, particularly to support private access to critical cloud-based business applications such as storage and videoconferencing.

“To ensure end-to-end IP connectivity to customers, we rely on Google Kubernetes Engine to keep at least one instance of the Layer 2 orchestration microservices working,” says Paul. “If, for example, the Layer 2 microservice fails, customers won't be

able share data and business applications between branch offices or access cloud services, thereby hampering their employees' ability to work remotely.”

To illustrate, when a user clicks on Console Connect and requests a connection to Google Cloud from Hong Kong to San Jose, PCCW Global uses Elasticsearch to record the API call to the workflow engine. The workflow engine microservice then talks to a microservice to create an interconnect on Google Cloud for an inbound enterprise user that wants to connect in San Jose and hand it off at this network-to-network interface. The final step is for PCCW Global to configure several devices across the network, both an edge switch and a border router, and then add routers to complete the transaction.

“Behind the scenes, it takes a sheer breadth of networks to orchestrate connectivity in minutes. Every click on Console Connect to create a point-to-point or point-to-multipoint connection could involve 10 to 20 different devices and systems,” says Paul. “Fortunately, we can deploy the open-source app Elasticsearch on Google Cloud to capture every single transaction across every single device and every single subsystem with a correlation ID.”

